



PHANTOM 4.8 // DIAGRAM 05 — CAMERA DISCOVERY & STREAMING LIFECYCLE

GUJARAT POLICE DEPARTMENT • NETRAM COMMAND & CONTROL

ACTIVE C2
EXHIBIT

1. DYNAMIC SENTINEL CATALOGUE INGESTION (/api/ingest)

Corp8SourceAdapter Dynamic Sync

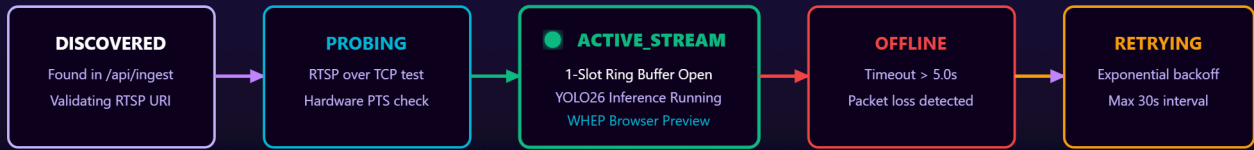
- Zero hard-coded cameras; polls external registry
- Reconciles camera state: Active, Inactive, Maintenance

Sync cycle: Every 60s background async loop

5-Stage Exponential Backoff on Failure

On HTTP 502 / Connection Timeout:
2s -> 4s -> 8s -> 16s -> 30s (Max)
Enters DEGRADED state; preserves cached GIS data

2. CAMERA STREAMING STATE MACHINE & PROTOCOL RESOLUTION



3. ADAPTIVE LOAD PACING & BANDWIDTH ALLOCATION PROFILES

PROFILE: LOW

Bitrate: 500 kbps
Framerate: 5 FPS
Resolution: 720p

OPERATIONAL USE:

- Rural highway stretches
- Low-activity perimeters
- Zero watchlist presence

COMPUTE SAVINGS:

83% GPU cycle reduction
Background polling only

PROFILE: MEDIUM

Bitrate: 1500 kbps
Framerate: 15 FPS
Resolution: 1080p

OPERATIONAL USE:

- Standard municipal grid
- Moderate urban traffic
- Baseline ANPR active

DEFAULT STATE:

48 streams per L40S
Nominal state-wide profile

PROFILE: HIGH

Bitrate: 2500 kbps
Framerate: 25 FPS
Resolution: 1080p

OPERATIONAL USE:

- Major highway tolls
- Dense city junctions
- Active manual viewing

HIGH PRECISION:

Full ANPR + 2W/Car
Fast-moving traffic

BURST_TRACKING

Bitrate: 4000 kbps
Framerate: 30 FPS
Resolution: 1080p HQ

TRIGGER CONDITIONS:

- Watchlist match hit
- Suspect vehicle pursuit
- VIP escort corridor

AUTO-DEESCALATION:

Reverts to MEDIUM after
120s of no target sighting

TECHNICAL SPECIFICATIONS & ARCHITECTURE MECHANISMS:

- Zero hard-coded streams: SentinelCatalogueService dynamically polls and reconciles cameras from /api/ingest via Corp8 adapter.
- 5-stage exponential backoff (2s -> 4s -> 8s -> 16s -> 30s) prevents cascading gateway failures during network blips.
- Dynamic Load Pacing Profiles: LOW (500 kbps), MEDIUM (1500 kbps), HIGH (2500 kbps), BURST_TRACKING (4000 kbps).